March 2025

Econ 2042: Statistics for Economists

Tutorial Exercise II:

1). Three weekly English newspapers: The Monitor (A); Addis Tribune (B); and The Reporter (C) are popular in Addis. A sample of 100 readers gives the number of persons reading one, two or three weeklies:

$$\begin{aligned} N &= 100; \; n(A) = 70; \quad n(B) = 85; \quad n(C) = 65 \\ n(A \cap B) &= 25; \; n(A \cap C) = 35; \; n(B \cap C) = 30; \; n(A \cap B \cap C) = 15 \end{aligned}$$

Find the number of persons who read

a). A only.

b). At least two weeklies.

c). B and C only.

2). There are two identical boxes containing: box 1 contains 4 white and 3 red balls, and box 2 contains 3 white and 7 red balls, respectively. A box is chosen at random and a ball is drawn.

a). What is the probability that the drawn ball is white?

b). If the drawn ball is white, what is the probability that it came from the first box?

3). A bolt factory has two machines, A and B, producing bolts in the ratio 60 to 40. The proportion of defective bolts produced by A is 0.05 and that of B is 0.04. If a bolt has been found to be defective, what is the probability that it was produced by:

a). Machine A, and

b). Machine B.

4). How many permutations are possible for the letters of the following words:

a). Punctuation

b). Ethiopian

5). In how many ways can 4 boys occupy their seats in a local bus, which has only 6 vacant seats?

6). How many 4 digit numbers can be formed by using the digits 1 to 9, if the digit once selected

a). Cannot be repeated?

b). Can be repeated?

7). In how many ways can 7 persons be seated at a round table?

8). If you have 6 friends, in how many ways can you invite one or more to your party?

9). Obtain the number of ways in which 10 girls and 10 boys can form a circle so that no two girls are together.

10). There are 10 candidates for a scholarship in Economics, 15 for 2 in Mathematics and 12 for one in Physics. In how many ways can the scholarships be awarded?

11). A gentleman invites a party of 19 friends to tea and places 10 of them at one round table. Show that the number of ways in which they can be arranged among themselves is ${}_{19}C_{10} \times (10-1)!$.

12). From 6 male teachers and 4 female teachers a committee of 4 is to be formed. In how many ways can this be done if the committee contains

a). at least one female,

b). at least one male,

c). at least one of each?

13). Show that the number of ways in which n distinct books can be arranged on a shelf so that two particular books shall not be together is $(n-2) \times (n-1)!$

14). In a certain competitive examination you can score $0, 1, 2, 3, 4$ or 5 grade points in one attempt. If you make 7 attempts, what is the total number of ways of scoring 30 grade points in aggregate?

15). Fourteen men are to be grouped into three different committees each investigating a different subject and each consisting of at least four members. In how many ways can the men be distributed if four specified men are to serve in the same committee?

16). What is the total number of ways in which 10 different rings can be bought by 8 different ladies when all rings must be bought and there is no compulsion on any lady to buy?

17). If A' is the complement of A, show that $\Pr(A') = 1 - \Pr(A)$.

18). If A and B are any two events and A' and B' are their complements, respectively, show that

a). $\Pr(A) = \Pr(A \cap B) + \Pr(A \cap B')$, and

b). $\Pr(B) = \Pr(A \cap B) + \Pr(A' \cap B)$.

19). Show that

a). $(A \cup B)' = A' \cap B'$ and

b). $(A \cap B)' = A' \cup B'$.

20). Show that

$$\Pr(A \cap B) \leq \Pr(A) \leq \Pr(A \cup B) \leq \Pr(A) + \Pr(B)$$

21). Show the following result

$$\Pr(A \cap B) = \Pr(A) - \Pr(A') + 1 - \Pr(A \cup B)$$

22). If an unbiased die is tossed n times, what is the probability of obtaining 6 at least once?

23). A glass bottle is used to carry milk every day. If the probability of its being broken on any particular day is $1/1001$, what is the probability that it will survive for four days?

24). What is the probability of obtaining 'double six' at least once if two unbiased dice are thrown n times in succession?

25 Show that the number of throws required to assume a probability of more than $\frac{1}{2}$ of obtaining double six at least once in successive throws of two unbiased dice is 25.

26). If an unbiased die is thrown 5 times in succession, what is the probability of obtaining a six on at least two consecutive occasions?

27). Twelve persons are seated at a round table. What is the probability that there are exactly 3 persons between A and B?

28). Three boxes marked I, II and III have the following contents:

Box I–one white, two black and three red balls,

Box II–two white, one black and one red ball,

Box III–four white, five black and three red balls.

One of the boxes has been selected at random and the two balls drawn are found to be white and red. What is the probability that they were drawn from box II?

29). A survey of female students at the Addis Ababa University shows that 50% of them come from Addis, 25% from Dire Dawa, 15% from Bahir Dar and 10% from Mekele. It is observed that 80% of those from Addis, 50% of those from Dire Dawa, 5% of those from Bahir Dar and 1% of those from Mekelle wear jeans.

If a female student has been selected at random and is found wearing jeans, what is the probability that she comes from Addis?

30). The probability that A and B speak the truth independently are p and p', respectively. If they make the same statement, what is the probability that the statement is, in fact, true?